

Sri Lanka Institute of Information Technology



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SE2030 – Software Engineering.

WEB BASED MUSIC STORE
Group – Y2-S1-MLB-B4G1-04

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1. Introduction

1.1 Project Background

MelodyMart is a web-based digital music store that allows customers to browse, search, purchase, and download digital music, including individual songs and full albums. The system is designed to give music lovers a convenient and secure online shopping experience while giving administrators an efficient platform to manage music content, listener accounts, artist information, sales, and promotional activities.

Traditional music retail is limited by physical stock, store opening hours, and manual record keeping. Melody Mart removes these limitations by centralising every part of the music-buying journey from discovery to purchase to download into a single, always-available web platform.

1.2 Problem Statement

Customers who want to buy digital music today often struggle with fragmented platforms, poor search functionality, and inconsistent purchasing and download experiences. At the same time, store administrators lack a unified system to manage the music catalogue, track sales, support artists, and respond to customer needs. MelodyMart addresses both problems by providing one integrated platform for customers and administrators alike.

1.3 Project Objectives

- Develop a secure, web-based platform for purchasing digital music.
- Provide a simple, user-friendly interface for browsing and buying songs and albums.
- Simplify the process of purchasing and downloading digital music.
- Enable administrators to efficiently manage the music catalogue, listeners, artists, and orders.
- Generate sales reports and dashboard analytics to support business decisions.
- Build a scalable and maintainable system that can support future growth.

1.4 Target Users

- Customer / Listener
- Artist
- Sales Manager
- Customer Support

1.5 Scope

The MelodyMart system will include the following within its scope:

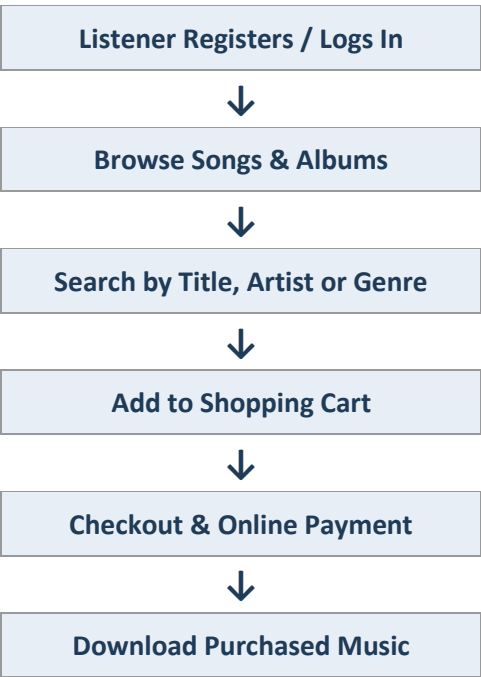
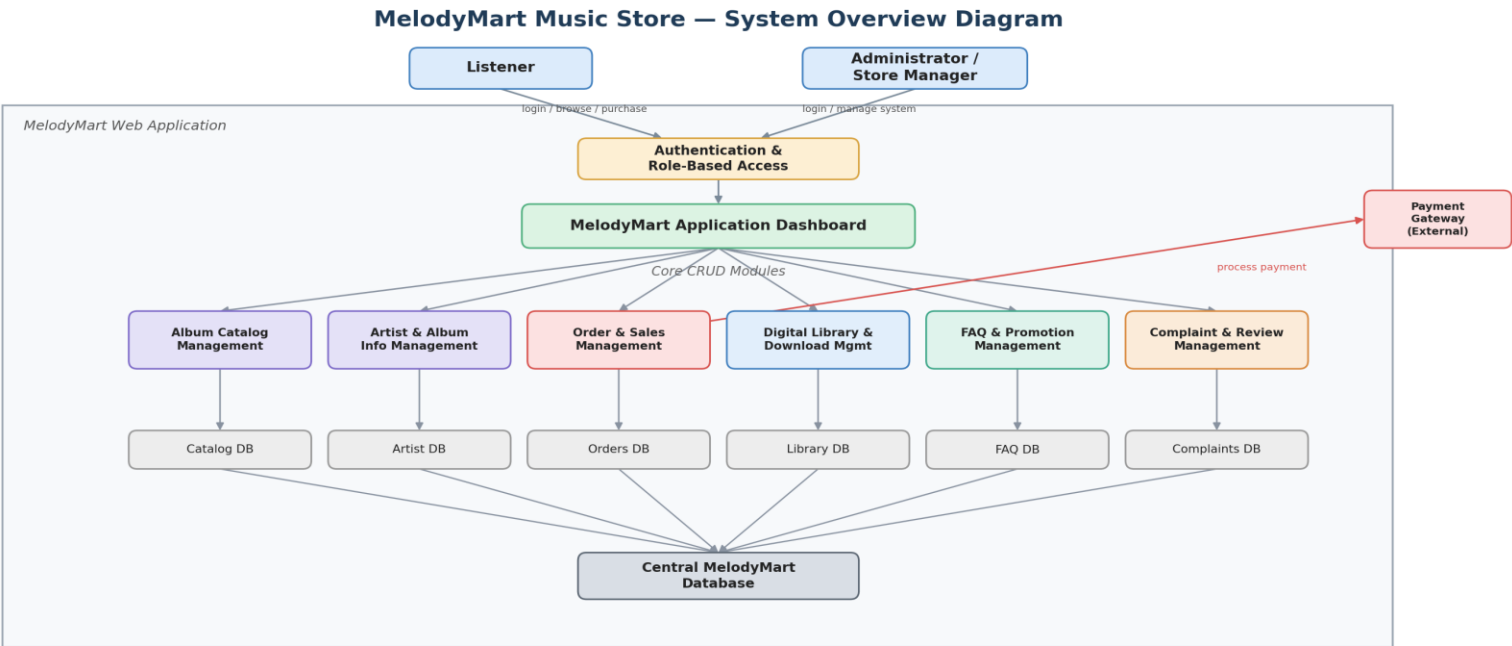
- Customer registration and login
- Browsing and searching songs and albums by title, artist, or genre
- Shopping cart and checkout
- Online payment processing
- Downloading purchased music
- Artist and album artwork management
- Sales reports and dashboard analytics

1.6 Limitations

- No live streaming of music.
- No physical CD or merchandise sales.
- An active internet connection is required to use the system.
- Payments are processed only through the integrated online payment gateway.

2. System Overview Diagram

The diagram below illustrates the typical journey a listener takes through the MelodyMart system, from registration to downloading purchased music



3. Functional Requirements

Functional requirements describe what the MelodyMart system must be able to do.

ID	Requirement
01	The system shall allow listeners to register a new account.
02	The system shall allow listeners to log in and log out.
03	The system shall allow listeners to update their profile information.
04	The system shall allow listeners to browse songs and albums.
05	The system shall allow listeners to search music by title, artist, or genre.
06	The system shall allow listeners to add albums to a shopping cart.
07	The system shall allow listeners to purchase music through an online payment gateway.
08	The system shall allow listeners to download purchased songs and albums.
09	The system shall allow listeners to view their purchase history.
10	The system shall allow listeners to contact customer support.
11	The system shall allow administrators to manage the album catalog.
12	The system shall allow administrators to manage artist and album information.
13	The system shall allow administrators to manage orders and generate sales reports.
14	The system shall allow administrators to manage the digital library and download settings.
15	The system shall allow administrators to manage FAQs and promotions.
16	The system shall allow administrators to manage listener complaints and reviews.
17	The system shall allow listeners to submit complaints and reviews about their purchases.

4. Non-Functional Requirements

4.1 Performance

- Pages shall load within 3 seconds under normal network conditions.
- Search results shall be returned quickly, even with a large music catalogue.
- The system shall support multiple concurrent users without noticeable slowdown.

4.2 Security

- User passwords must be encrypted.
- Only authorised users may access administrator features (role based authorisation).
- Online payments must be processed securely through a trusted payment gateway.

4.3 Reliability

- The system shall perform automatic backups of critical data.
- The system shall support data recovery in the event of failure.

4.4 Availability

- The system shall be available 24/7, subject to scheduled maintenance.

4.5 Usability

- The system shall have a responsive design that works across desktop and mobile devices.
- Navigation shall be simple and intuitive for non-technical users.

4.6 Maintainability

- The system shall be built with a modular architecture to simplify future updates.
- Source code shall be well-documented.

4.7 Scalability

- The system shall support a growing number of users, songs, and albums.
- The architecture shall allow for future feature integration.

5. Major Stakeholders / Users

The table below lists the key stakeholders identified during requirement gathering, together with their role in the MelodyMart system.

Stakeholder	Category	Role / Responsibility
Listener (Customer)	End user	Browses, purchases, and downloads music.
Store Manager	Internal user	Manages the overall website, catalogue, and operations.
Artist	External user	Provides music content and profile information.
Sales Executive	Internal user	Monitor sales performance.
Customer Support	Internal user	Resolves listener issues and enquiries.
Payment Gateway Provider	External user	Processes secure online payments.
Music catalogue coordinator	Internal user	Add new songs, updates album details and maintains the music catalogue.
Business owner	Internal user	Oversees the entire business and makes important decision.

6. Six Major Functions

6.1 Album Catalog Management (Wijesinghe B.T.A.B)

Who uses it: Administrator

What it does: Manages the catalog of individual albums available in the MelodyMart store, including adding new albums and maintaining accurate album details.

Expected outcome: The store maintains an accurate, well-organised, and up-to-date catalog of albums for listeners to browse and purchase.

CRUD	Operations
Create	Add new album
Read	View album list/details
Update	Edit album information
Delete	Delete album

6.2 Artist & Album Information Management (Sneha N)

Who uses it: Administrator

What it does: Maintains detailed information about artists and their albums, including biographies, genres, and album metadata.

Expected outcome: Listeners always see accurate, current artist and album information across the store.

CRUD	Operations
Create	Add new artist/album info
Read	View artist & album information
Update	Update artist or album details
Delete	Delete artist or album record

6.3 Order & Sales Management (Gajaweera G.A.S.N)

Who uses it: Listener, Administrator, Sales Manager

What it does: Handles the placing of orders by listeners and lets administrators track order status, sales performance, and generate sales reports.

Expected outcome: Listeners can complete purchases smoothly, and the business gains clear visibility into sales performance.

CRUD	Operations
Create	Create new order
Read	View orders & sales reports
Update	Update order status/details
Delete	Delete/cancel order

6.4 Digital Library & Download Management (Divyanjalee P.N)

Who uses it: Listener, Administrator

What it does: Manages the digital music library that listeners download from after purchase, while administrators upload files and control download settings.

Expected outcome: Listeners can reliably access and download the music they have purchased, and the digital library stays organised and up to date.

CRUD	Operations
Create	Upload digital music/files
Read	View library & download history
Update	Update file or download settings
Delete	Delete digital files

6.5 FAQ & Promotion Management (Shagavan S.K)

Who uses it: Administrator, Marketing Team

What it does: Manages the FAQs and promotional complains shown to listeners on the store.

Expected outcome: Listeners have access to helpful information and current promotions, kept accurate and up to date by administrators.

CRUD	Operations
Create	Add FAQ or promotion
Read	View FAQs & promotions
Update	Edit FAQ or promotion
Delete	Remove FAQ or promotion

6.6 Complaint & Review Management (Thisaranga B.M.A.T)

Who uses it: Listener, Customer Support, Administrator

What it does: Allows listeners to submit complaints and reviews about their purchases, and lets customer support and administrators manage and resolve them.

Expected outcome: Listener concerns are tracked and resolved efficiently, and feedback helps the business improve its products and service.

CRUD	Operations
Create	Submit complaint/review
Read	View complaints & reviews
Update	Update complaint status or review
Delete	Delete complaint or review

7. Minor Functions

In addition to the major functions above, MelodyMart provides the following supporting features:

- Login / Logout
- Forgot Password / Password Reset
- Change Password
- Email Verification and Notifications
- Search Listeners (administrator)
- Filter Songs by Genre
- Music Preview
- Upload Profile Picture
- View Account Status
- Dashboard Statistics and Top Selling Songs Display
- Contact Us / Contact Support
- Wishlist

8. System Limitations / Constraints

- An active internet connection is required to use the system.
- The system is designed to work on modern web browsers.
- Only digital music is sold no physical products are offered.
- Only online payments are supported no offline or cash payments.
- Songs become downloadable only after successful payment is confirmed.

9. Simple Timeline

The table below outlines the team's planned activities from Week 3 to Week 14.

Week	Activity
3	Requirement gathering & proposal
4	UI design & database design
5	Use case & class diagrams
6	Album catalog module
7	Artist & album information module
8	Order & sales management module
9	Digital library & download module
10	Progress presentation
11	Testing & bug fixes
12	FAQ, promotion, complaint & review modules final documentation
13	Final presentation & viva
14	Buffer & improvements

10. Conclusion

The MelodyMart Music Store Management System is designed to provide a secure, efficient, and user-friendly platform for both listeners and administrators. By bringing together album catalog management, artist and album information management, order and sales management, digital library and download management, FAQ and promotion management, and complaint and review management into one integrated system, MelodyMart simplifies both the customer's music buying journey and the store's day-to-day operations.

For listeners, the system offers a convenient way to discover, purchase, and download music securely, backed by a responsive and easy to navigate interface. For administrators, it offers the tools needed to manage content, track sales, and make informed business decisions through dashboard analytics and reporting.

With a clear scope, well defined functional and nonfunctional requirements, and a realistic development timeline, the team is confident in its plan to design, build, and deliver a scalable and maintainable MelodyMart system that meets the needs of all identified stakeholders.